A Report on

4-Element Circular Microstrip Patch Antenna Array

for

EWRS Minor-Project of Third Year, (TY Sem-VI)

in

Electronics & Telecommunication Engineering

by

B-625 Atharva Jadhav

B-607 Deevy Bhanushali

Under the guidance of

Dr. Suman Wadkar

Mahatma Education Society's

Pillai College of Engineering

(Autonomous)

Accredited A+ by NAAC

Dr. K. M. Vasudevan Pillai Campus,

Plot No. 10, Sector-16, New Panvel – 410206

A.Y. 2025-26

CERTIFICATE

This is to certify that the project entitled 4-Element Circular Microstrip Patch Antenna Array is a Bonafide work of

1. Atharva Jadhav
2. Deevy Bhanushali

Submitted to the University of Mumbai in partial fulfilment of the requirement for the award of **EWRS Minor-Project of Third Year, (TY Sem-VI)** in **Electronics & Telecommunication Engineering** as laid down by **University of Mumbai** during academic year **2025-26.**

Dr. Suman Wadkar

Supervisor

ABSTRACT

A 4-element circular microstrip patch antenna array is designed and analyzed for operation at **2.5 GHz**. The antenna consists of four circular radiating patches arranged in a linear configuration and connected using a corporate feed network to ensure equal power distribution.

The use of an antenna array significantly improves gain, directivity, and radiation efficiency compared to a single patch antenna. Circular patches are chosen due to their symmetrical radiation characteristics and ease of design.

The antenna is fabricated on an **FR-4 substrate** and simulated using **CST Studio Suite**. Key performance parameters such as **return loss (S11), voltage standing wave ratio (VSWR), gain, and radiation pattern** are analyzed.

The results show good impedance matching with S11 less than -10 dB at the desired frequency and VSWR close to unity. The antenna also exhibits directional radiation with improved gain.

This design is suitable for applications in **Wi-Fi, ISM band communication, Bluetooth devices, and wireless communication systems**, offering a compact, low-cost, and efficient solution.

LIST OF FIGURES

Fig No.	**Name of Figure**	Page No.
Figure 3.2	CST Simulation Model 4-Element Circular Microstrip Patch Antenna Array	11
Figure 3.3	S1.1 Graph	11
Figure 3.4	Radiation Pattern (Far-Field Directivity)	12
Figure 3.5	Smith Representation of S11 Parameter Chart	12
Figure 3.6	VSWR Graph	13

INDEX

CONTENTS		
Certificate	……	i
Abstract	……	ii
List of Figures	……... ……	iii
1.	Introduction…… 1.1 Objectives	6
2.	Software Used……	7
3.	Methodology ……. 3.1 System Design Approach 3.2 Simulated Outputs 3.3 Development Method	8
4.	Results and Discussion…….…….. 4.1 Simulated Results 4.2 Tested Results 4.3 Advantages 4.4 Disadvantages 4.5 Applications	13
5.	Conclusions…….	15
References	……..	15

1.Introduction

Microstrip antennas are widely used in modern wireless communication systems due to their compact size, low profile, light weight, and ease of fabrication. These antennas consist of a metallic patch printed on a dielectric substrate with a ground plane on the opposite side. However, a single patch antenna suffers from limitations such as low gain and narrow bandwidth. To overcome these issues, antenna arrays are used. An antenna array consists of multiple radiating elements arranged in a specific configuration to improve gain and directivity.

In this project, a **4-element circular microstrip patch antenna array** is designed for **2.5 GHz operation**. The circular patches provide better symmetry and stable radiation characteristics. A corporate feed network is used to distribute power equally among all patches.

The antenna is designed and simulated using CST Studio Suite, and its performance is analyzed in terms of return loss, VSWR, and radiation pattern.

1.1 Objectives

CST Studio Suite is used for electromagnetic simulation of antennas. It provides accurate 3D modeling and analysis of RF components using numerical techniques like FIT.

Features include:

- 3D modeling
- S-parameter analysis
- Radiation pattern visualization
- Optimization tools

2.Software Used

CST Studio Suite is a powerful electromagnetic simulation software used for designing, analyzing, and optimizing high-frequency components such as antennas, microwave circuits, and RF systems. It is widely used by engineers and researchers for accurate 3D modeling and simulation of electromagnetic fields.

CST uses advanced numerical techniques such as the Finite Integration Technique (FIT) and Time Domain Solver to analyze the behavior of electromagnetic waves. It allows users to design complex antenna structures, including microstrip and E-shaped antennas, and evaluate their performance before physical fabrication.

Features of CST Studio Suite:-

- 3D modeling of antenna structures
- Accurate electromagnetic field simulation
- Time domain and frequency domain analysis
- Easy visualization of results (graphs and radiation patterns)
- Optimization tools for improving design performance

In this project, CST is used to design and simulate the Y slot microstrip antenna. The antenna geometry is created by defining the substrate, patch, slots, coaxial feedline and ground plane. After setting the required parameters and boundary conditions, simulations are performed to analyze key performance metrics such as:

- Return Loss (S11)
- Voltage Standing Wave Ratio (VSWR)
- Gain
- Radiation Pattern
- Bandwidth

The simulation results help in understanding the antenna performance and optimizing the design parameters for better efficiency and bandwidth.

Advantages of Using CST

- Reduces the need for physical prototyping
- Saves time and cost
- Provides accurate and reliable results
- User-friendly interface

3.Methodology

3.1 System Design Approach

Design Parameters:

$$f = 2.5\,GHz$$
$$\varepsilon_r = 4.4$$
$$h = 1.6\,mm$$

Wavelength Calculation

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{2.5 \times 10^9} = 120\,mm$$

Effective Wavelength

$$\lambda_{eff} = \frac{120}{\sqrt{4.4}} = 57.2\,mm$$

Patch Spacing

$$d = \frac{\lambda}{2} = 60\,mm$$

Patch Radius Calculation

$$F = \frac{8.791 \times 10^9}{f\sqrt{\varepsilon_r}} = 16.77\,mm$$
$$a = \frac{F}{\sqrt{1 + \frac{2h}{\pi F \varepsilon_r}\left[\ln\left(\frac{\pi F}{2h}\right) + 1.7726\right]}}$$
$$a \approx 16.27\,mm$$

Final Radius (Optimized)

$$a = 15.4\,mm$$

3.2 Simulated Outputs

Steps to Simulate Y slot Microstrip Antenna in CST:

1. Open CST Studio Suite and create a new project. Select Microwave and RF and choose Time Domain Solver.
2. Set the units for the project such as millimeter for length and GHz for frequency.
3. Define the substrate by creating a brick. Assign FR-4 material and set its dimensions according to design.
4. Create the ground plane on the bottom side of the substrate using a metallic (copper) layer.
5. Design the rectangular patch on top of the substrate using copper material.
6. Modify the patch by cutting two parallel slots and one horizontal slot to form the Y shaped structure.
7. Add the coaxial feed line to excite the antenna and connect it properly to the patch.
8. Define the port (waveguide port or discrete port) at the feed point for signal input.
9. Set boundary conditions as open (add space) to simulate free space radiation.
10. Set the frequency range for simulation (for example around 2.45 GHz).
11. Run the simulation using the solver.
12. After simulation, observe results such as S11 (return loss), VSWR, gain, and radiation pattern.
13. Optimize the design by adjusting dimensions like slot size, feed position, and patch size to improve performance.

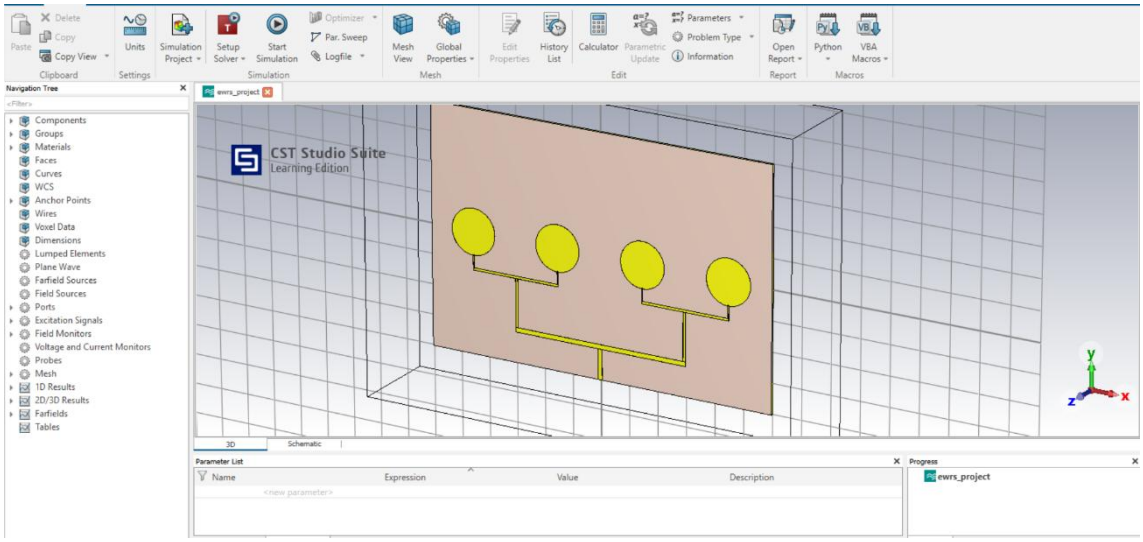


Figure 3.2 - CST Simulation Model of 4-element circular microstrip patch antenna array

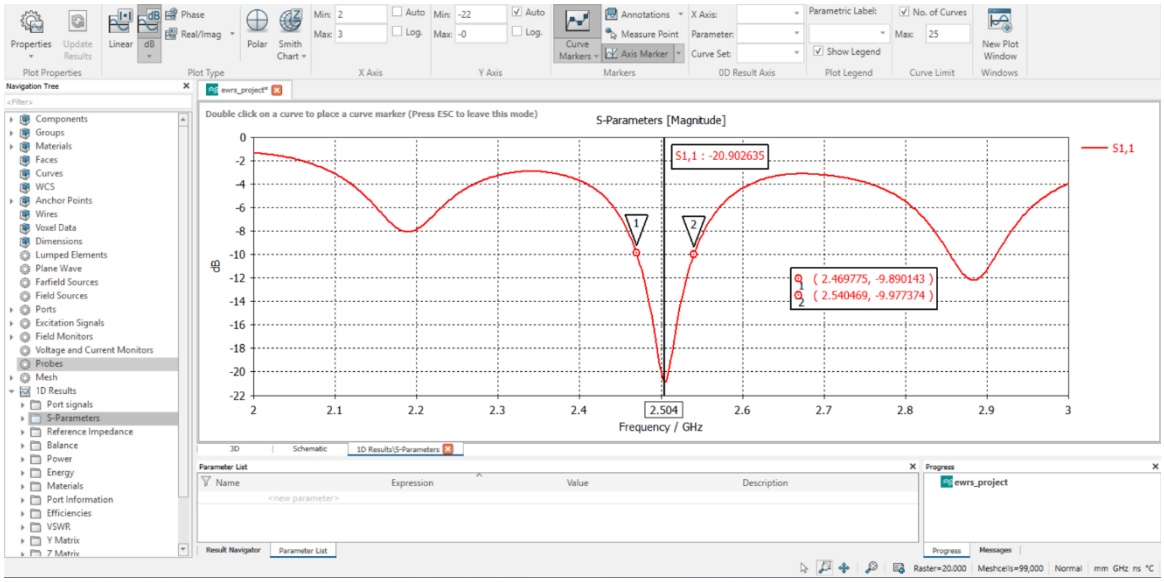


Figure 3.3 – S1.1 Graph

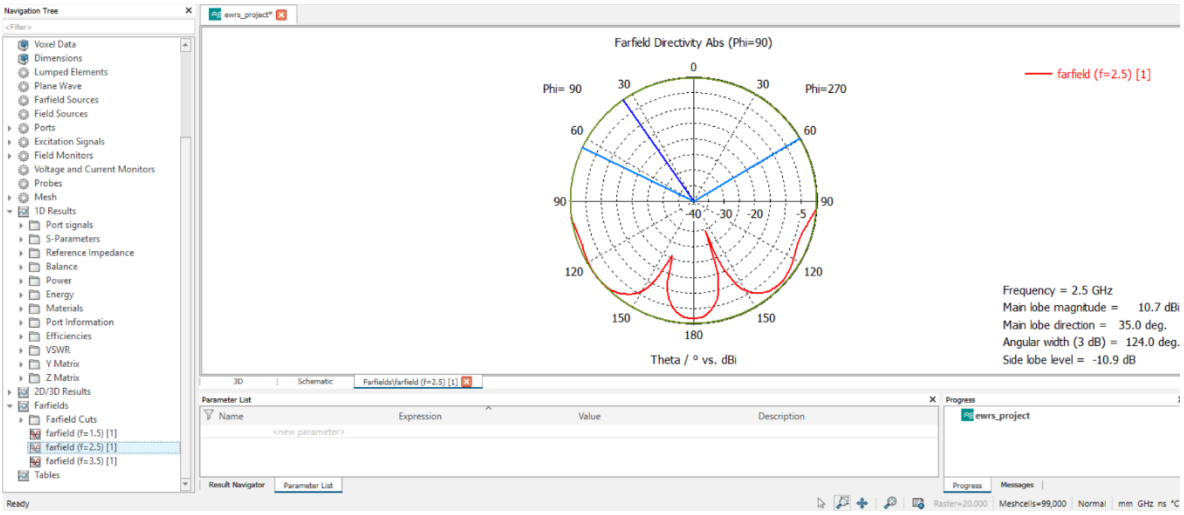


Figure 3.4 – Radiation Pattern (Far-Field Directivity)

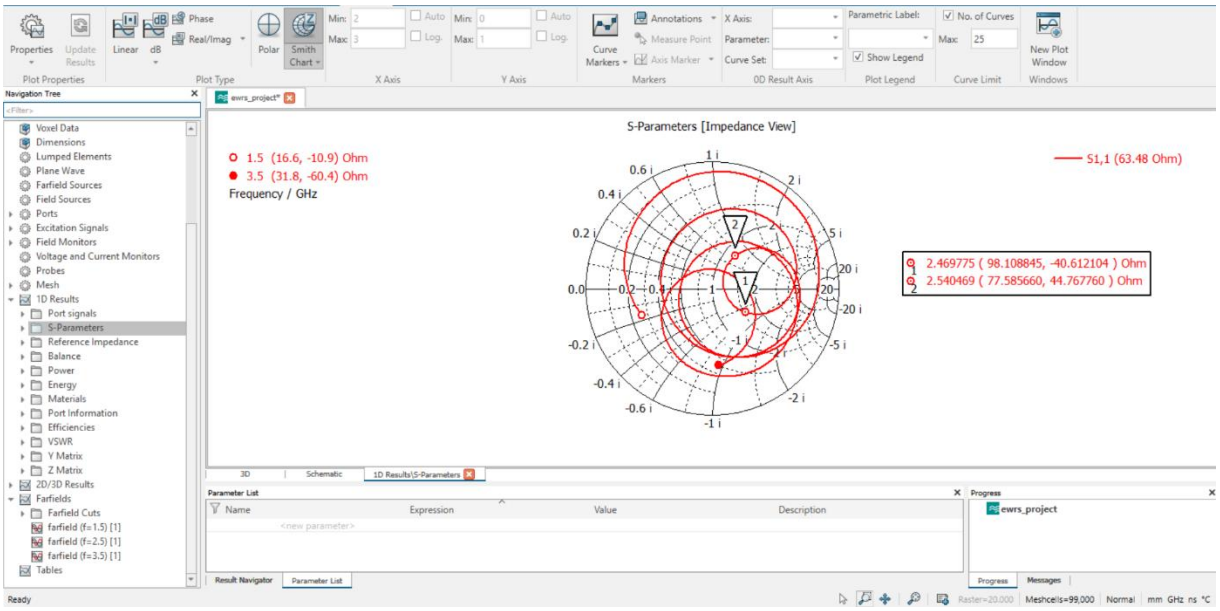


Figure 3.5 – Smith Representation of S11 Parameter Chart

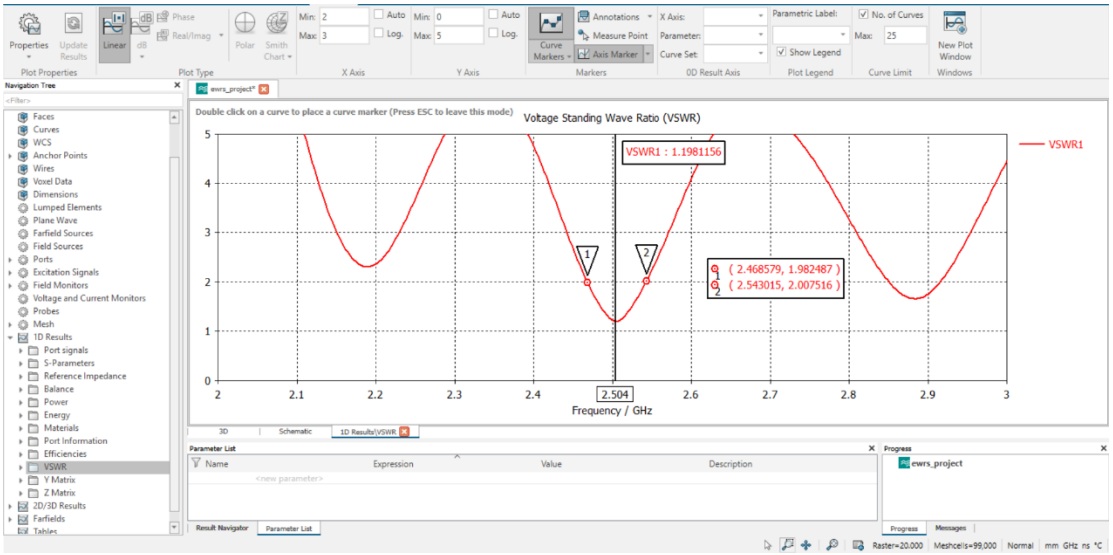


Figure 3.6 – VSWR Graph

4.3 Advantages

- **High Gain:**
  The use of a 4-element array significantly increases gain compared to a single patch antenna.
- **Improved Directivity:**
  The antenna radiates more power in a specific direction, improving signal strength.
- **Good Impedance Matching:**
  The antenna achieves low return loss (S11 < -10 dB), ensuring efficient power transfer.
- **Low VSWR:**
  VSWR close to 1 indicates minimal signal reflection and better performance.
- **Compact and Lightweight:**
  Microstrip antennas are easy to fabricate and require less space.
- **Ease of Fabrication:**
  The antenna can be easily manufactured using standard PCB techniques.
- **Suitable for Integration:**
  Can be easily integrated with RF circuits and wireless systems.

4.4 Disadvantages

- **Complex Feed Network:**
  The corporate feed network increases design complexity.
- **Dielectric Losses:**
  FR-4 substrate introduces losses, reducing efficiency slightly.
- **Limited Bandwidth:**
  Microstrip antennas generally have narrow bandwidth.
- **Fabrication Sensitivity:**
  Small changes in dimensions can affect antenna performance.
- **Mutual Coupling:**
  Interaction between patches can slightly affect radiation characteristics.
- **Beam Tilt Issue:**
  Improper feed alignment may cause shift in radiation direction.

4.5 Applications

- **Wi-Fi Systems (2.4–2.5 GHz band)**
- **Wireless Communication Systems**
- **WLAN (Wireless Local Area Networks)**
- **Bluetooth Devices**
- **IoT (Internet of Things) Applications**
- **Radar and Surveillance Systems**
- **Satellite Communication (low frequency bands)**

Conclusion

The **4-element circular microstrip patch antenna array** was successfully designed and simulated for operation at **2.5 GHz** using CST Studio Suite. The antenna demonstrates improved performance in terms of gain, directivity, and impedance matching compared to a single patch antenna.

The simulated results show that the antenna achieves **low return loss (S11 < -20 dB)** and **VSWR close to 1**, indicating efficient power transmission. The radiation pattern is directional, and the gain is significantly enhanced due to the array configuration.

Although the design introduces some complexity in the feed network and minor losses due to the FR-4 substrate, the overall performance is satisfactory for practical applications.

The antenna is well suited for **Wi-Fi, WLAN, and other wireless communication systems** operating in the 2.5 GHz band, making it a compact, efficient, and cost-effective solution.

References

[1] C. A. Balanis, *Antenna Theory: Analysis and Design*, 4th ed. Hoboken, NJ, USA: Wiley, 2016.

[2] D. M. Pozar, “Microstrip antennas,” *Proceedings of the IEEE*, vol. 80, no. 1, pp. 79–91, Jan. 1992.